Abdul Rehman

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Summary

Graduate with Master's in Intelligent Systems Engineering from Indiana University Bloomington. I have experience in systems research and software development ranging from firmware to cloud computing. My notable contributions are gVNIC driver development, UEFI graphics driver development, NVMe virtualization, and my recent work on custom Linux CPU scheduling for a serverless platform. I am proficient in Rust, C/C++, eBPF and Python development.

Work Experience

Indiana University Bloomington
Bloomington, IN, USA

Research Assistant — *Aug 2022 - May 2026*

Advisor: Prateek Sharma

- Serverless platform driven CPU loadbalancing (Linux, eBPF, SchedExt, Rust)
  - A CPU scheduling approach (using SchedExt framework) that uses function invocation frequency to assign scheduling domains.
  - https://abrehman94.github.io/projects/schedext_based_scheduling.html
- Contributed to the development of a serverless platform written in Rust (publication).
  - https://abrehman94.github.io/projects/iluvatar.html

Google
Sunnyvale, CA, USA

Software Development Intern — *May 2025 - Aug 2025*

- Development: Userspace poll mode driver for gVNIC (Google Virtual NIC) on Google Cloud Platform VMs.
- **Skills:** C++, VFIO-PCI, Linux Kernel, Test driven development

Siemens Industry Software Inc.
Mobile, AL, United States

Senior Software Engineer - Hypervisor Team — *Mar 2021 - Sep 2021*

- Siemens Hypervisor support on Intel Embedded Processors Elkhartlake (2021).
  - Release critical bug-fixes: ACPI parser, AHCI Virtualization, NVMe Virtualization
  - **Skills:** C/C++, ACPI, AHCI, NVMe, Type-1 virtualization

Mentor Graphics a Siemens Business
Lahore, Pakistan

Senior Software Engineer - Hypervisor Team — *Jan 2020 - Mar 2021*

- Feature Release: virtualized UEFI interface
  - Design and development of a non-volatile variable caching infrastructure to avoid SMI generation and provide real time gurantees for Guest RTOS.
  - **Skills:** C/C++, UEFI (Unified Extensible Firmware Interface), RTOS
- Feature Release: NVMe Virtualization
  - Performance improvement of NVMe virtualization infrastructure from 700 MB/s to 1.5 GB/s. Improved the infrastructure to process requests across SMP cores.
  - **Skills:** C/C++, Performance engineering

Software Engineer - Hypervisor Team — *Aug 2016 - Jan 2020*

- Feature Release: virtualized UEFI interface
  - Implementation of UEFI boot support for Guest OS (Windows, Linux, RTOS) on Siemens Type 1 Hypervisor.
  - Design and development of a UEFI driver for Intel Graphics Device (IGD) to allow early graphics for Linux and Windows guests.
  - **Skills:** C/C++, Intel graphics device
- Feature Release: NVMe Virtualization
  - Design and development of the infrastructure. Specifically, interrupt handling (PCI MSI-X capability) and I/O queue segregation for the virtual devices and hardware backend.
  - Adapting the Linux NVMe driver to act as a Paravirtualized Client for testing purpose.
  - **Skills:** C/C++, PCI, Virtualization
- Feature Releases for Mentor Embedded Hypervisor (Type 1)
  - Design and development of VT-d DMAR/IOMMU driver to allow device isolation and memory remapping of guests.
  - **Skills:** C, x86 IOMMU, VT-d

Software Engineer - Embedded UI Graphics Team — *Jun 2015 - Aug 2016*

- Feature Release: Qt GUI Framework version 5.4 on Nucleus RTOS
  - Porting of the framework based on a previous work for Qt 4.0.
  - Performance optimization of the ported framework.
- **Skills:** C, Qt, Nucleus RTOS

Publications

[1] Alexander Fuerst, **Abdul Rehman**, Prateek Sharma. Ilúvatar: A Fast Control Plane for Serverless Computing. *High-Performance Parallel and Distributed Computing(HPDC) '23*, Acceptance Rate 21%

Education

Indiana University Bloomington *Bloomington, United States*

MSc in Intelligent Systems Engineering *Aug 2022 - May 2026*

- CGPA 3.67/4.00
- **Courses:** Compilers, Operating Systems, Applied Algorithms, Cloud Computing, Signal Processing using Machine Learning, Deep Learning

National University of Sciences and Technology *Islamabad, Pakistan*

Electrical Engineering *Aug 2011 - June 2015*

- CGPA 3.92/4.00
- **Courses:** Embedded Systems, Digital System Design, Digital Signal Processing

Awards

2023	**Travel Grant**, High-Performance Parallel and Distributed Computing 2023	*Orlando, USA*
2016	**Appreciation Certificates: exceptional debugging skills, high quality work**, Mentor a Siemens Business	*Lahore, Pakistan*

Skills

Programming Languages	Rust, C/C++, Bash, Python
Application Level	eBPF, Distributed Systems (serverless, InfluxDB), Containerization(Docker, containerd)
Close to hardware	Linux Kernel, Type 1 Hypervisors (ACRN/XEN/MEHV/Siemens), UEFI Driver, ACPI, x86, Lauterbach Trace-32 Debuggers
Machine Learning	Pandas, NumPy
Tools	tmux, vim, ssh